Chapter 6

6. Clustering methods.

6.1. K–Means Clustering.

6.2. Hierarchical Clustering.

Introduction

Clustering refers to techniques for finding subgroups (clusters) in data. These subgroups should be formed such that within a cluster data behave similarly. In contrast to classification there is no response involved that should be predicted. Hence, there is also no way of cross-validation and there is no a priori knowledge (no training sample) what we are looking for.

Clustering falls under the generic term of unsupervised learning.

Examples:

- An online shop may try to find groups of people with similar browsing and purchase history in order to advertise products efficiently.
- A search engine can display specific results to people with a similar search pattern.

K-means clustering

Suppose we want group $X_1, \ldots, X_n$ in K clusters. Then, generally speaking, we search for a partition $C_1, \ldots, C_K$ of the index set $\{1, \ldots, n\}$ and define $X_{C_k} = \{X_i : i \in C_k\}$ to be the k-th cluster.

The sets C_k should be chosen such, that the *within-cluster variation*, let us denote by $V(X_{C_k})$, is as small as possible.

Hence denoting $\mathcal{P}_{n,K}$ the set of partitions of $\{1, \ldots, n\}$ into K subsets, we seek for

$$\underset{\{C_1, \ldots, C_K\} \in \mathcal{P}_{n,K}}{\operatorname{argmin}} \left\{ \sum_{k=1}^{K} V(X_{C_k}) \right\}.$$

K-means clustering

We need to define the within-cluster variation.

A common measure is the *squared Euclidian distance*:

$$V(X_{C_k}) := \frac{1}{|C_k|} \sum_{i,j \in C_k} \|X_i - X_j\|^2.$$

Writing $\|X_i - \bar{X}_k + \bar{X}_k - X_j\|^2$, where $\bar{X}_k = \sum_{i \in C_k} X_i$, it can be easily shown that

$$V(X_{C_k}) := 2 \sum_{i \in C_k} \|X_i - \bar{X}_k\|^2.$$

K-means clustering

The minimization problem above is highly complex and numerically very costly unless n and K are small. Here is an algorithm which works well and is fast:

1. Randomly assign a number in $1, \ldots, K$ to each observation. All observations which got the number k are assigned to cluster k.
2. Compute the mean vector of each cluster.
3. Check for each observation which of the cluster means is closest (in Euclidian distance) and assign it to this cluster.
4. If all observations stay in their current cluster, stop. Otherwise go back to step 2.

K-means clustering

Remarks:

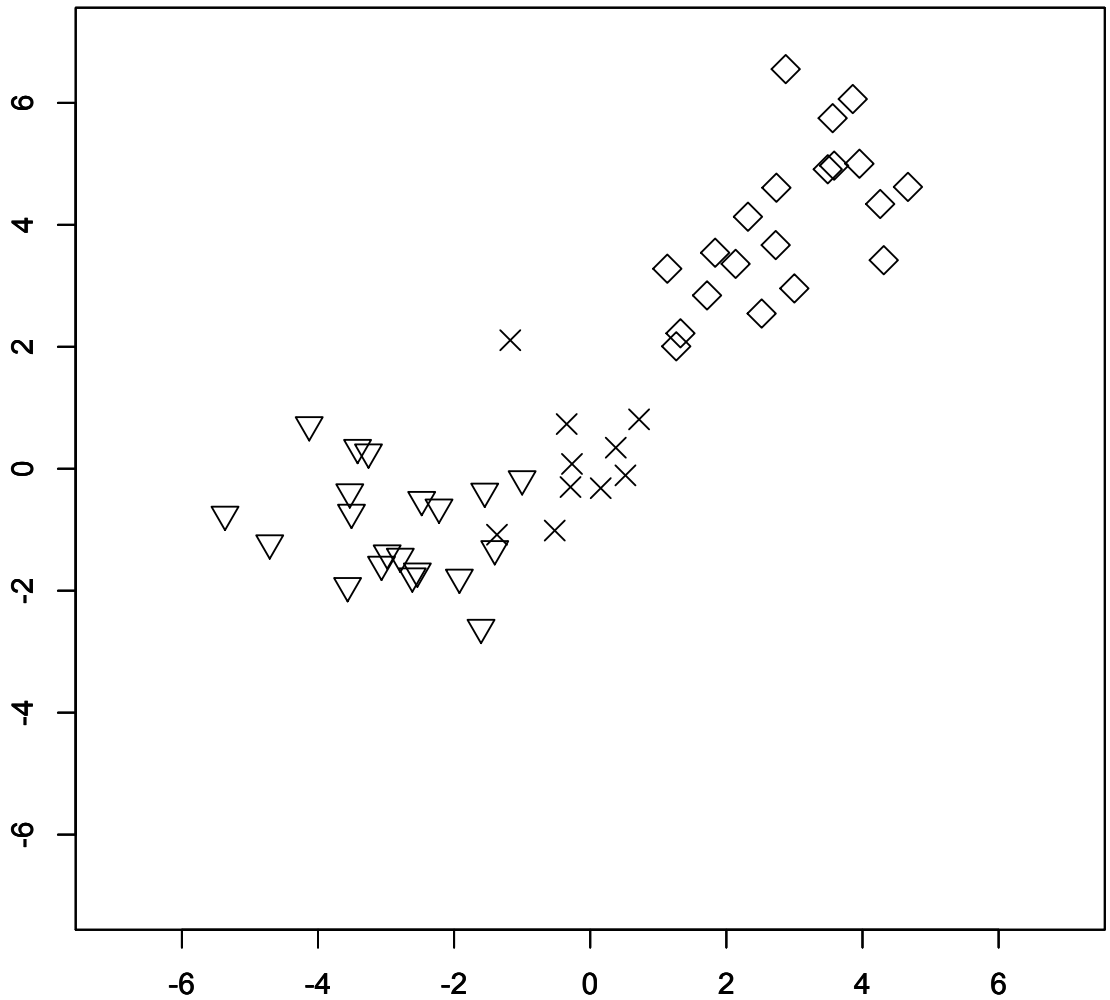
- In each step the sum of within-cluster variations decreases.
- As soon as the algorithm stops, a local minimum has been reached. It is not necessarily the global minimum. Therefore it is important to run the algorithm serval times and choose the best solution.

K-means clustering

We give an illustration in R. First we generate artificial data coming from 3 different multivariate normal populations.

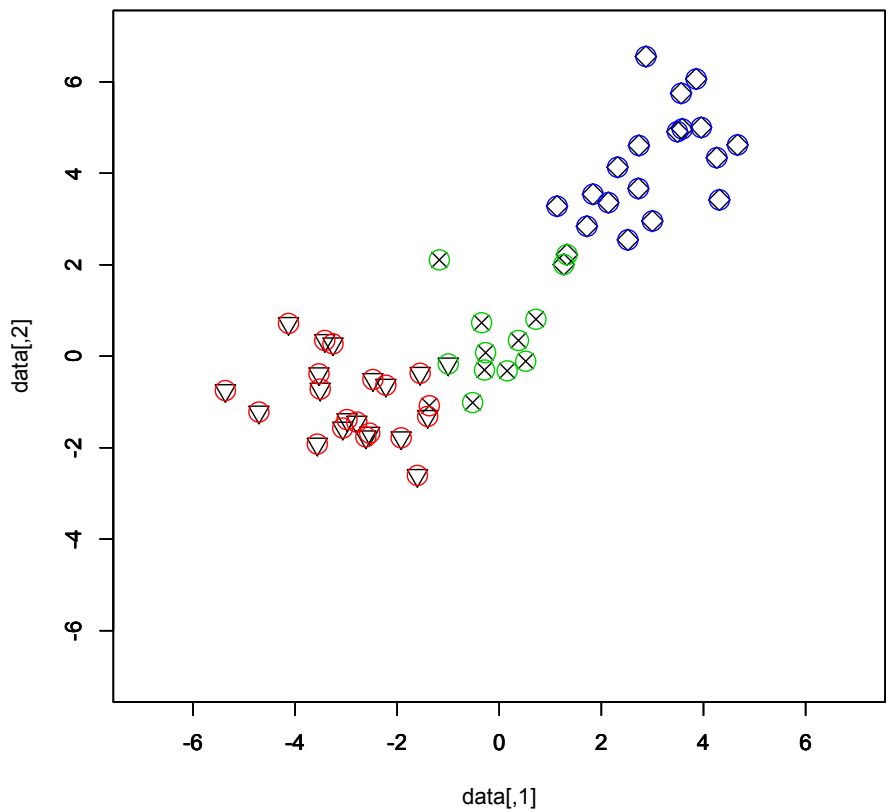
```
> library(mvtnorm)
> c1=rmvnorm(10,mean=c(0,0),sigma=diag(2)*0.5)
> c2=rmvnorm(20,mean=c(3,4),sigma=matrix(c(2,1,1,2)*0.5,ncol=2))
> c3=rmvnorm(20,mean=c(-3,-1),sigma=matrix(c(2,-1,-1,2)*0.5,ncol=2))
>
> data = rbind(c1,c2,c3)
>
> plot(c1,pch=4,xlab="",ylab="",xlim=c(-7,7),ylim=c(-7,7),cex=1.5)
> par(new=TRUE)
> plot(c2,pch=5,xlab="",ylab="",xlim=c(-7,7),ylim=c(-7,7),cex=1.5)
> par(new=TRUE)
> plot(c3,pch=6,xlab="",ylab="",xlim=c(-7,7),ylim=c(-7,7),cex=1.5)
```

K-means clustering



K-means clustering

```
> km=kmeans(data,3,iter.max=5,nstart=20)
> par(new=TRUE)
> plot(data,col=(km$cluster+1),pch=1,cex=2,xlim=c(-7,7),ylim=c(-7,7))
```

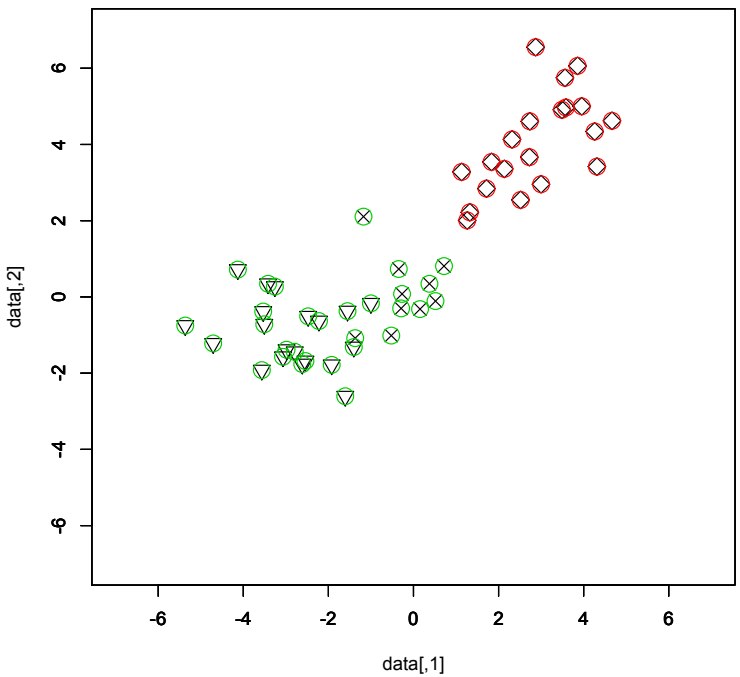


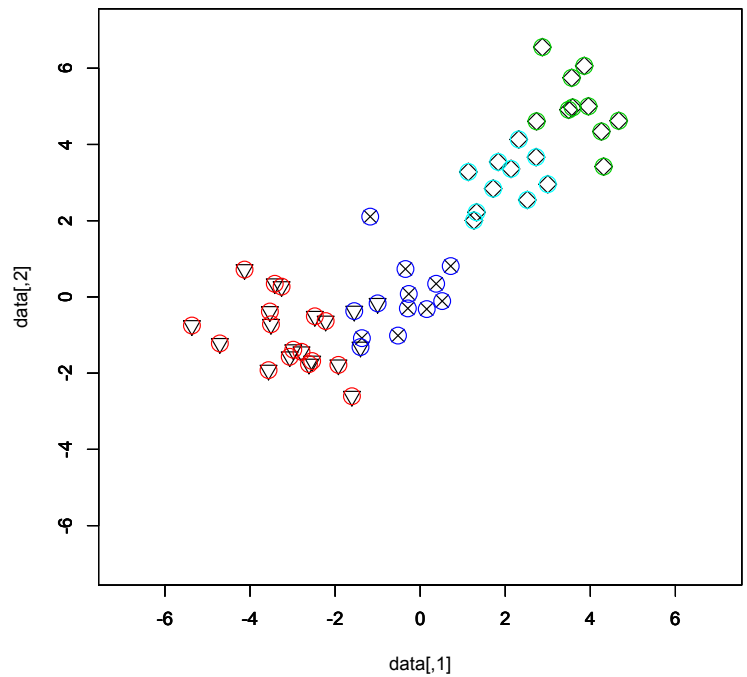
K-means clustering

Remarks:

- Here `nstart` is the number of initial (random) configurations that will be checked. In order to obtain a global minimum (or at least close to global) a large values are recommended. Especially when the clusters are not so clear cut, the results may vary.
- The second argument is the number of clusters that should be considered.
- `iter.max` is the maximum number of iterations the algorithm will be run. Convergence is usually quite fast.
- In practice we don't know K. On the next slide we show what happens when we use the same data and choose $K = 2$ or $K = 4$.

K-means clustering





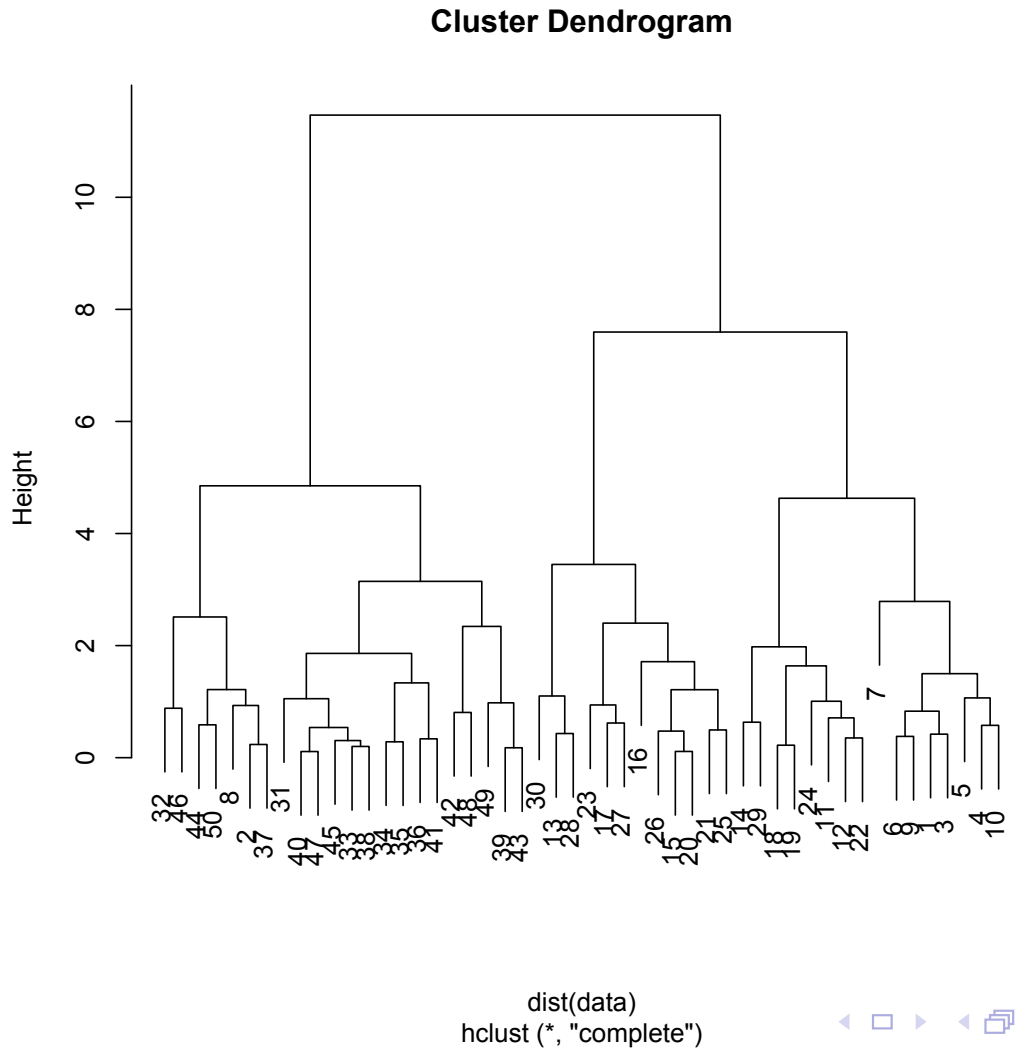
Hierarchical Clustering

The K-means clustering requires to pre-specify the number of clusters. Hierarchical clustering doesn't require such tuning and gives a simple graphical tool—the dendrogram—to find clusters. Any number of clusters can be extracted from one analysis.

Let us look at the data from above. And start with a plot of the dendrogram.

Hierarchical Clustering

```
> hc=hclust(dist(data),method="complete")
> plot(hc)
```



Hierarchical Clustering

The graph above shows the dendrogram from a bottom up clustering with complete linkage and Euclidian distance. We are going to explain these options later. Let us first explain how the dendrogram is interpeted.

1. Each leaf of the dendrogram represents one of the 50 observations.
2. As we move up the tree some leaves begin to fuse into branches, which correspond to similar observations.
3. If we continue to move up further, brunches or leaves further fuse. The earlier the fusion occurs, the more similar the groups of observations are. The later, the more different.
4. The height of the fusion indicates, how different observations are.

Hierarchical Clustering

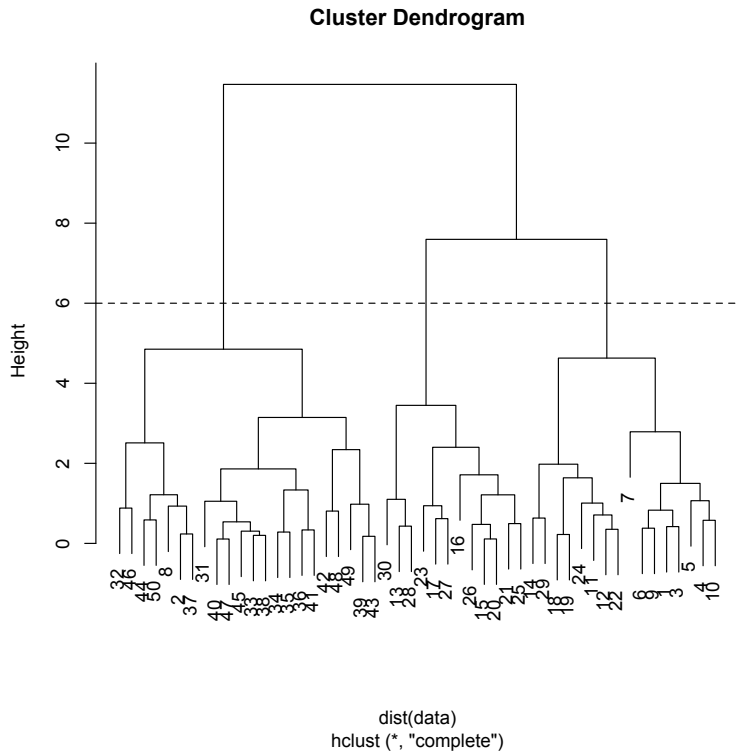
Remark. Note that in the above example observation 16 cannot be said to be closer to 26 as to 25. The horizontal distance plays no role. Whenever a fusion occurs, the position of the two branches can be swapped, without changing the meaning of the dendrogram.

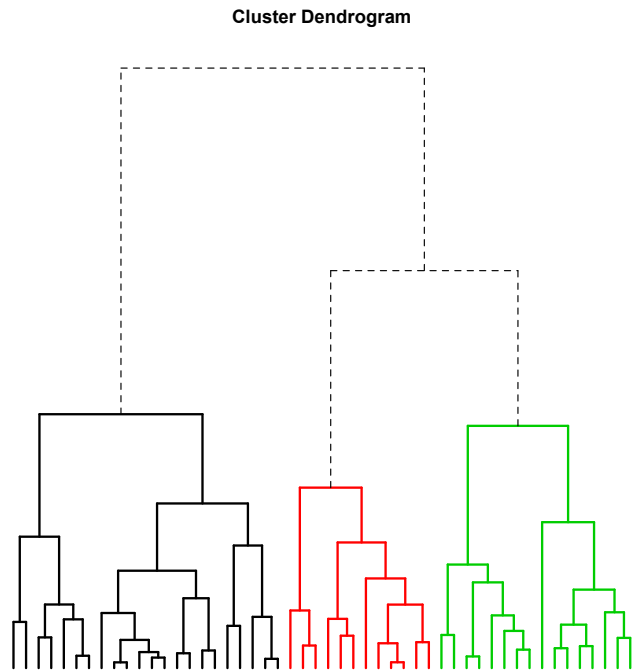
How do we obtain the clusters? We draw a horizontal line at some height. The entire branch beneath each cutting point forms one cluster. The lower the horizontal line is drawn the more clusters we get.

By progressively descending the cutting line we can get any number of clusters between 1 and n.

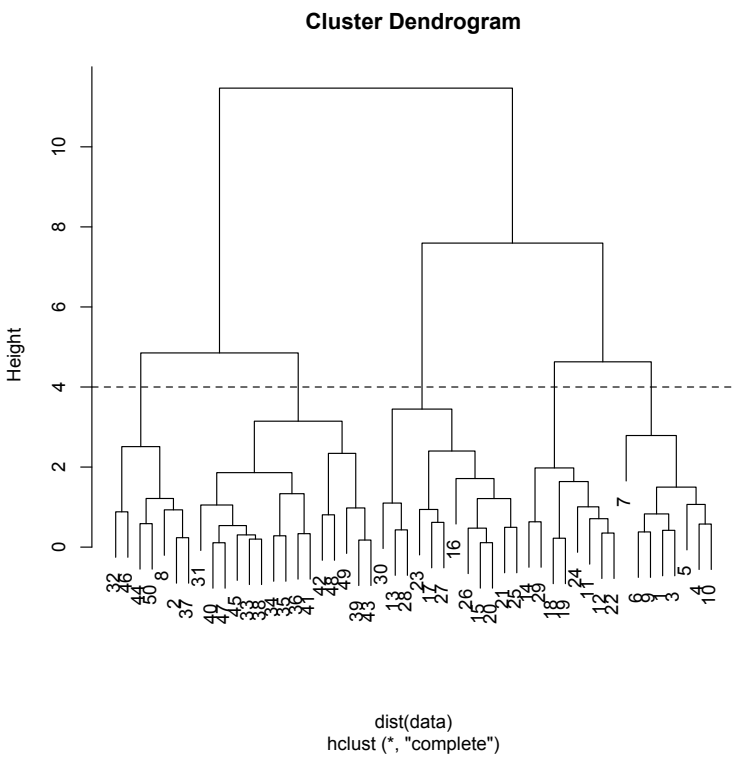
Hierarchical Clustering

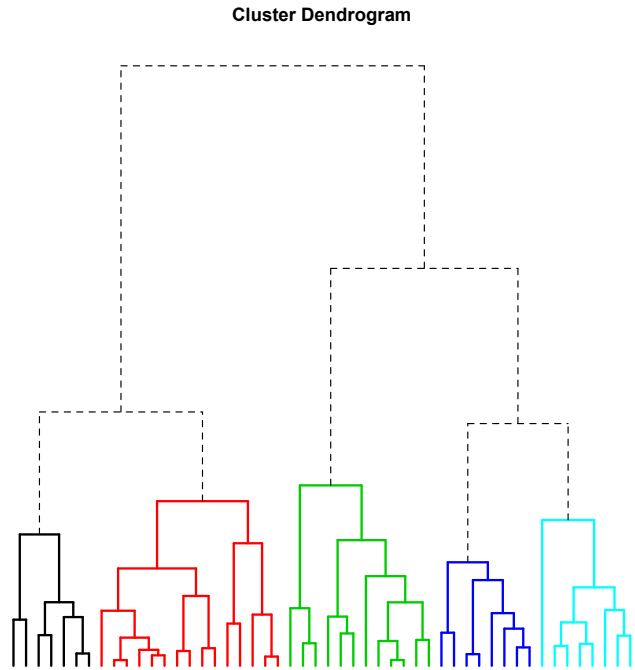
```
> source("http://addictedtor.free.fr/packages/A2R/lastVersion/R/code.R"
> A2Rplot(hc, k = 3, boxes = FALSE, col.down = 1:3)
```





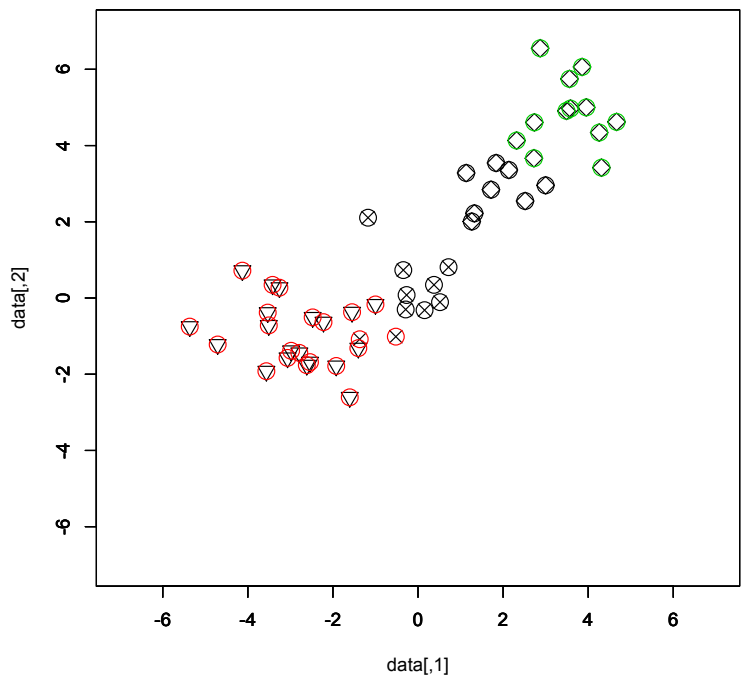
Hierarchical Clustering

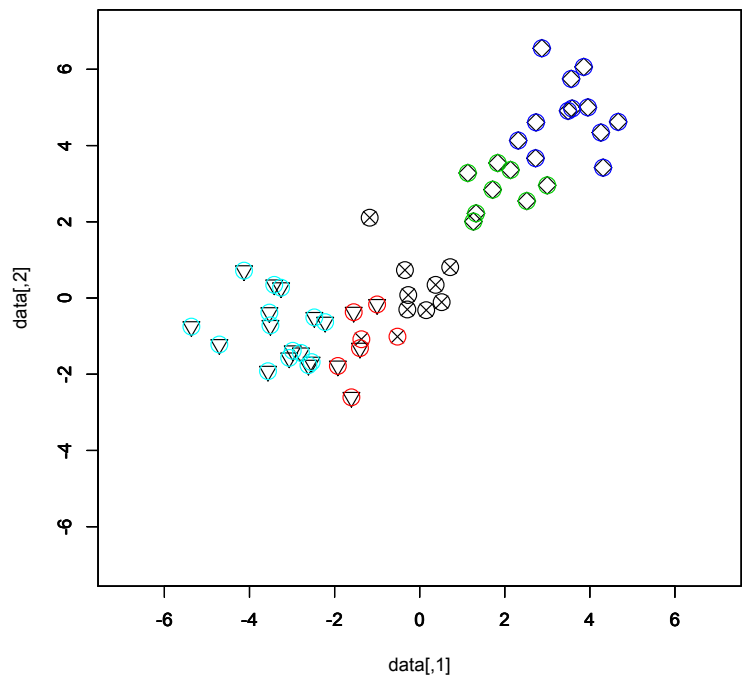




Hierarchical Clustering

```
> par(new=TRUE)
> plot(data,col=cutree(hc,3),pch=1,cex=2,xlim=c(-7,7),ylim=c(-7,7))
```





Hierarchical Clustering

Note that clusters obtained in this way are necessarily nested. This also why the procedure is named hierarchical clustering.

Note also that a hierarchical structure is not always realistic: if our data consist of hight and weight of people form Austria, USA and China, then two clusters may probably separate man and woman, whereas three clusters could distinguish the different countries. Here, apparently, we are not having a nested structure.

Hierarchical Clustering

This is the general form of the clustering algorithm:

1. Start with n clusters—1 observation per cluster.
2. Fuse the two most similar clusters $\Longrightarrow n-1$ clusters.
3. Proceed to further fuse two most similar clusters $\Longrightarrow n-2$ clusters.
4. ...
5. At the final step, all observations are in one cluster.

Hierarchical Clustering

How do we define the similarity/dissimilarity of clusters? This is defined as linkage. Here are four common choices:

complete	Compute all pairwise distances between observations in cluster A and cluster B and record the largest.
single	Compute all pairwise distances between observations in cluster A and cluster B and record the smallest.
average	Compute all pairwise distances between observations in cluster A and cluster B and record the average.
centroid	Compute distances between mean of cluster A and mean of cluster B.

As for distance: most common is to use Euclidian distance.

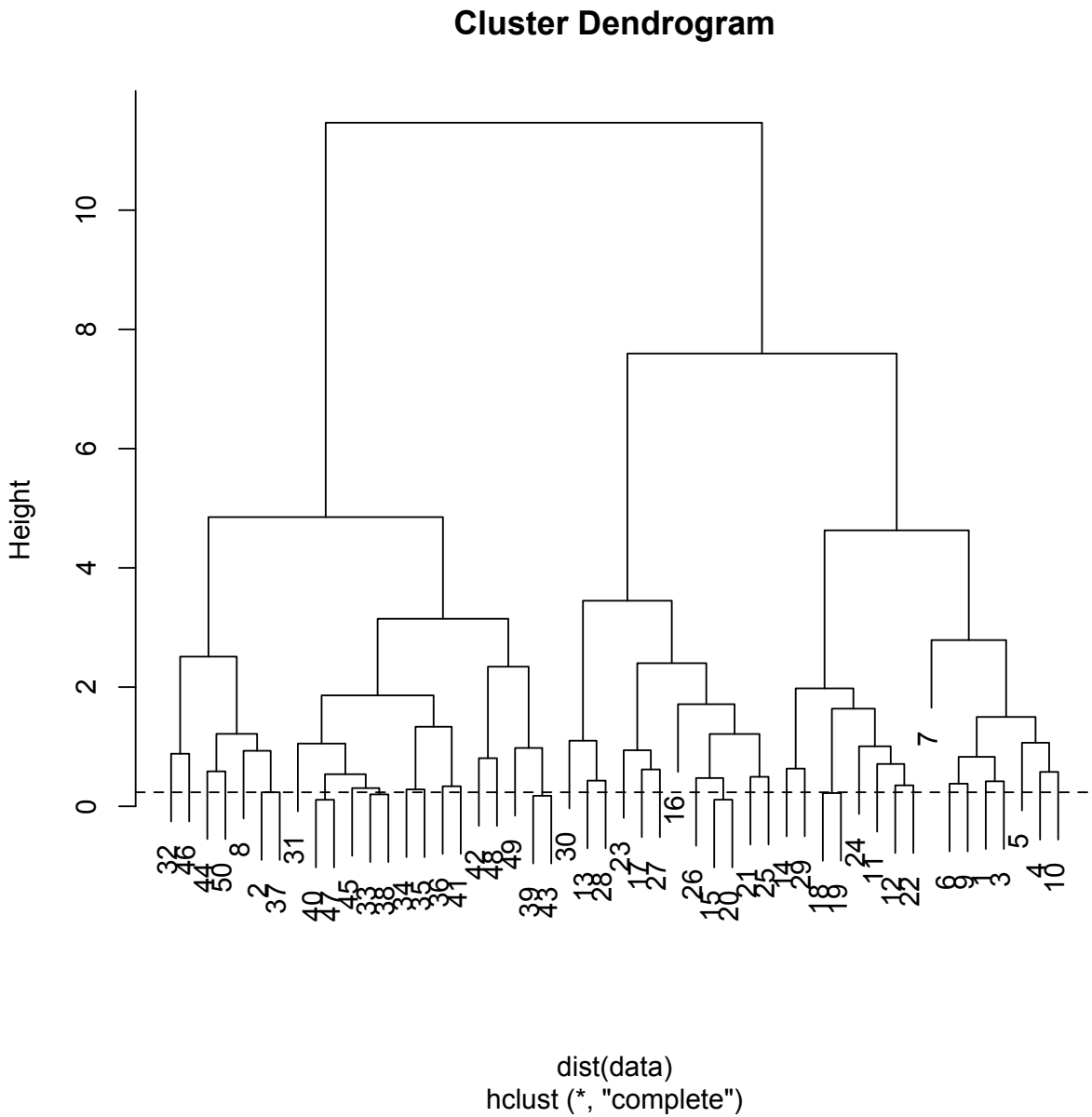
Hierarchical Clustering

The distance between clusters defines the hight of the dendrogram at which fusion takes place.

We illustrate this by computing the distance between the data points x_2 and x_{37} which are fused on bottom level (obs # 6 and 7 from left).

```
> plot(hc)
> abline(h=sqrt(sum((data[2,]-data[37,])^2)),lty=2)
```

Hierarchical Clustering



Remarks

Both clustering methods involve certain decision that have to be taken:

1. How many clusters are we looking for? Especially important for K-means.
2. Should we scale data?
3. What measure of similarity/dissimilarity (distance) are we using?
4. What linkage for hierarchical clustering should be used?

There is no clearcut answer to these question. The objective is to obtain interpretable results.

Real data illustration

We perform clustering on the track record data from Chapter 3.

```
> dat=read.table("T1-9.dat")
> data=dat[,2:7]
> rownames(data)=as.character(dat[,1])
> datas=scale(data)
> km=kmeans(datas,5,nstart=20)
> km
K-means clustering with 5 clusters of sizes 20, 12, 9, 2, 11

Cluster means:
          V2         V3         V4         V5         V6         V7
1 -0.2569842 -0.2314452 -0.2441374 -0.3960654 -0.4917828 -0.4712720
2 -1.0833372 -1.1411799 -1.1383690 -0.9773735 -0.8271171 -0.7286330
3  1.1221033  1.2239030  1.0946966  0.8931723  0.9240051  0.6505003
4  2.7714261  2.7571703  2.6936396  3.0227167  3.4165751  3.7153923
5  0.2270862  0.1630542  0.3003297  0.5059824  0.4192606  0.4439772

Clustering vector:
 ARG  AUS  AUT  BEL  BER  BRA  CAN  CHI  CHN  COL  COK  CRC  CZE  DEN  DOM
   5    2    1    1    5    1    2    5    2    5    4    3    2    1    3
 FIN  FRA  GER  GBR  GRE  GUA  HUN  INA  IND  IRL  ISR  ITA  JPN  KEN KORS
   1    2    2    2    1    3    1    3    1    1    5    1    1    1    5
KORN  LUX  MAS  MRI  MEX  MYA  NED  NZL  NOR  PNG  PHI  POL  POR  ROM  RUS
   3    3    5    3    1    5    1    1    1    3    5    2    1    2    2
 SAM  SIN  ESP  SWE  SUI  TPE  THA  TUR  USA
   4    3    2    1    1    5    5    1    2

Within cluster sum of squares by cluster:
[1] 12.876192  8.034112 14.663470  7.793614  9.981340
 (between_SS / total_SS =  83.2 %)
```

Real data illustration

Let us try a dendrogram with Mahalanobis distances as dissimilarity measure.

```
> d=matrix(0,nrow=54,ncol=54)
> for(i in 1:54){
for(j in 1:54){
d[i,j]=t(datas[i,]-datas[j,])%*%solve(cov(datas))%*%(datas[i,]-datas[j,])
}
}
> d=as.dist(d)
> hc=hclust(d=d)
> plot(hc)
```

Hierarchical Clustering

